

RYAN KELLY

SOFTWARE ENGINEER

ABOUT ME

Software engineer with hands-on experience building Go (Golang) backend systems, full-stack web applications, and concurrent TCP services. Skilled in delivering real-world software that solves practical problems — from campus-wide resource-sharing platforms to multi-client chat servers. Comfortable working with Linux, Docker, SQLite, and Git-based collaborative workflows. Passionate about open-source development, low-resource environments, and impactful technology.

CONTACT

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GitHub: github.com/Rkelly-dot
Location: Kisumu, Kenya

SKILLS

Languages

- Go (Golang)
- JavaScript
- SQL

Tools & DevOps

- Git & GitHub
- Docker / Compose
- Linux / Bash
- SQLite

Core Concepts

- Concurrency (goroutines)
- TCP Networking
- REST API Design
- Data Structures
- Smart Contracts

EDUCATION

Software Engineering Fellowship
Zone01 Kisumu · 2026 – 2027
Project-based, peer-learning programme (42 School methodology) covering systems, algorithms, web dev, and DevOps.

WORK EXPERIENCE

Software Development Apprentice | Zone01 Kisumu *March 2026 – Present*

- Build Go-based backend systems implementing low-level logic, B-tree structures, and complex data pipelines.
- Automate development environments with Docker & Docker Compose for reproducible, isolated builds.
- Manage version history across repositories using Git — triaging issues and maintaining a clean commit record.
- Collaborate on code reviews, debugging sessions, and iterative problem-solving in a peer-learning environment.

Coding Piscine — Intensive Selection | Zone01 Kisumu *Feb – March 2026*

- Completed a rigorous 4-week engineering selection solving complex daily challenges in Go.
- Demonstrated proficiency in algorithms, data structures, and systems design under strict time pressure.

PROJECTS

ElimuLocal — Campus Study Resource Platform github.com/Rkelly-dot/elimulocal

Full-stack Go web app enabling Kenyan university students to share study materials over campus Wi-Fi — no internet required.

- Built complete backend using Go's standard library: HTTP server, HTML templating, multipart file uploads/downloads, and SQLite persistence.
- Implemented search, university filtering, upvoting, and download counters; embedded local fonts for full offline operation.
- Roadmapped 3-month growth from MVP to multi-campus deployment with authentication and PDF preview.

NetCat — Multi-Room TCP Chat Server

Unix netcat-inspired TCP server supporting concurrent multi-client connections, rooms, and message history.

- Hub–Room–Client architecture using goroutines, channels, and mutexes for deadlock-free concurrent I/O.
- Features: timestamped messages, history replay (100 msgs), dynamic username/room switching, full unit test suite.

ADDITIONAL INFORMATION

Hackathon — Blockchain Donor-NGO Platform | Zone01 Kisumu · 48hr *2026*

- Built a blockchain transparency platform where smart contracts release funds to NGOs only after predefined objectives are verified — eliminating misuse.
- Integrated third-party blockchain layer and designed the full contribution-to-release funding flow from scratch.

Cursor Buildathon — Deployed App in 1hr | Cursor *2026*

- Designed, built, and deployed a fully functional application within a 1-hour limit — demonstrating rapid prototyping under extreme constraints.